Project File

Web Programming using

Python and JavaScript

Submitted By:

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Course: B.Tech CSE

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Submitted To:

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EXPERIMENT – 1

❖ **AIM:** Personal Portfolio Website using HTML, CSS Styling and JavaScript.

❖ **Tasks:**

- **Project Setup & Introduction –** Create Project folder and Basic linked files.
- **HTML Structure –** Build portfolio section using semantic HTML.
- **Styling with CSS –** Apply layout, colors, spacing, and neat design.
- **Projects & Skills Display –** Show skills and projects clearly on the page.
- **JavaScript Interactivity –** Add basic interactive functionality using JavaScript.
- **Contact Form –** Create a form with user submission feedback.

❖ **Tech Stack Used:**

- **HTML, CSS, and JavaScript**

❖ **CODES:**

➢ **HTML Code:**

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Aditya Solutions</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <header class="hero">
        <h1>Aditya Solutions</h1>
        <p>Empowering Businesses Through Digital Innovation</p>
        <a href="#contact" class="btn">Get Started</a>
    </header>
```

```html
    <section id="about">
        <h2>About Us</h2>
        <p>
            Aditya Solutions is a forward-thinking technology company delivering
            innovative digital solutions that help businesses grow, scale,
            and succeed in the modern digital world.
        </p>
    </section>

    <section id="services">
        <h2>Our Services</h2>

        <div class="services-container">
            <div class="service-card">
                <h3>Web Development</h3>
                <p>Modern, responsive, and high-performing websites tailored for
your brand.</p>
            </div>

            <div class="service-card">
                <h3>Mobile App Development</h3>
                <p>Powerful Android and iOS applications designed for seamless user
experience.</p>
            </div>

            <div class="service-card">
                <h3>Digital Marketing</h3>
                <p>Boost your visibility with strategic SEO and targeted marketing
campaigns.</p>
            </div>
        </div>
    </section>

    <section id="testimonials">
        <h2>What Our Clients Say</h2>

        <div class="testimonial">
            <p>"Aditya Solutions transformed our online presence completely."</p>
            <span>- Sarah Williams</span>
        </div>

        <div class="testimonial">
            <p>"Exceptional service, creative ideas, and on-time delivery."</p>
            <span>- James Anderson</span>
        </div>
    </section>

    <section id="contact">
        <h2>Contact Us</h2>

        <div class="contact-info">
            <p>Email: info@adityasolutions.com</p>
```

```html
            <p>Phone: +1 234 567 890</p>
            <p>Address: 123 Innovation Street, Silicon City</p>
        </div>

        <form class="contact-form">
            <input type="text" placeholder="Your Name" required>
            <input type="email" placeholder="Your Email" required>
            <textarea placeholder="Your Message" rows="4" required></textarea>
            <button type="submit">Send Message</button>
        </form>
    </section>

    <footer>
        <p>&copy; 2026 Aditya Solutions. All rights reserved.</p>
    </footer>

</body>
</html>
```

➢ CSS Code:

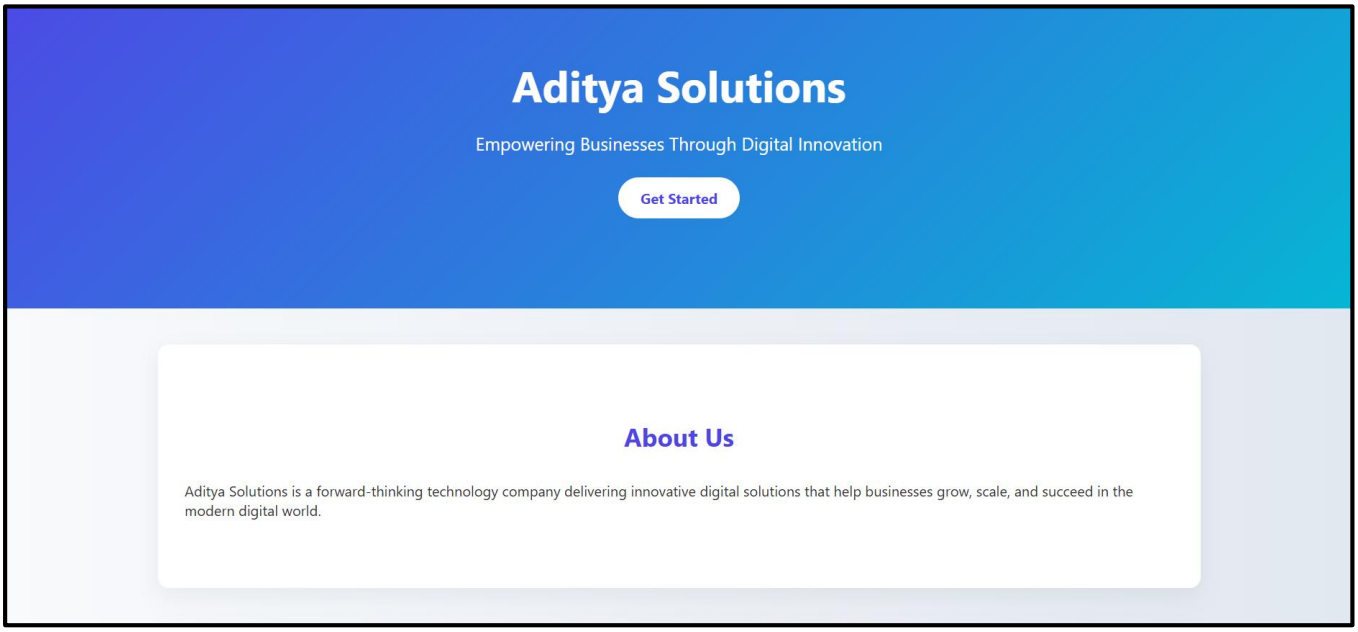
```css
body {
    font-family: 'Segoe UI', sans-serif;
    margin: 0;
    padding: 0;
    background: linear-gradient(to right, #f8fafc, #e2e8f0);
    color: #333;
    scroll-behavior: smooth;
}

.hero {
    background: linear-gradient(135deg, #4f46e5, #06b6d4);
    color: white;
    text-align: center;
    padding: 100px 20px;
}

.hero h1 {
    font-size: 48px;
    margin-bottom: 15px;
}

.hero p {
    font-size: 20px;
    margin-bottom: 25px;
}

.btn {
    display: inline-block;
    padding: 12px 25px;
```

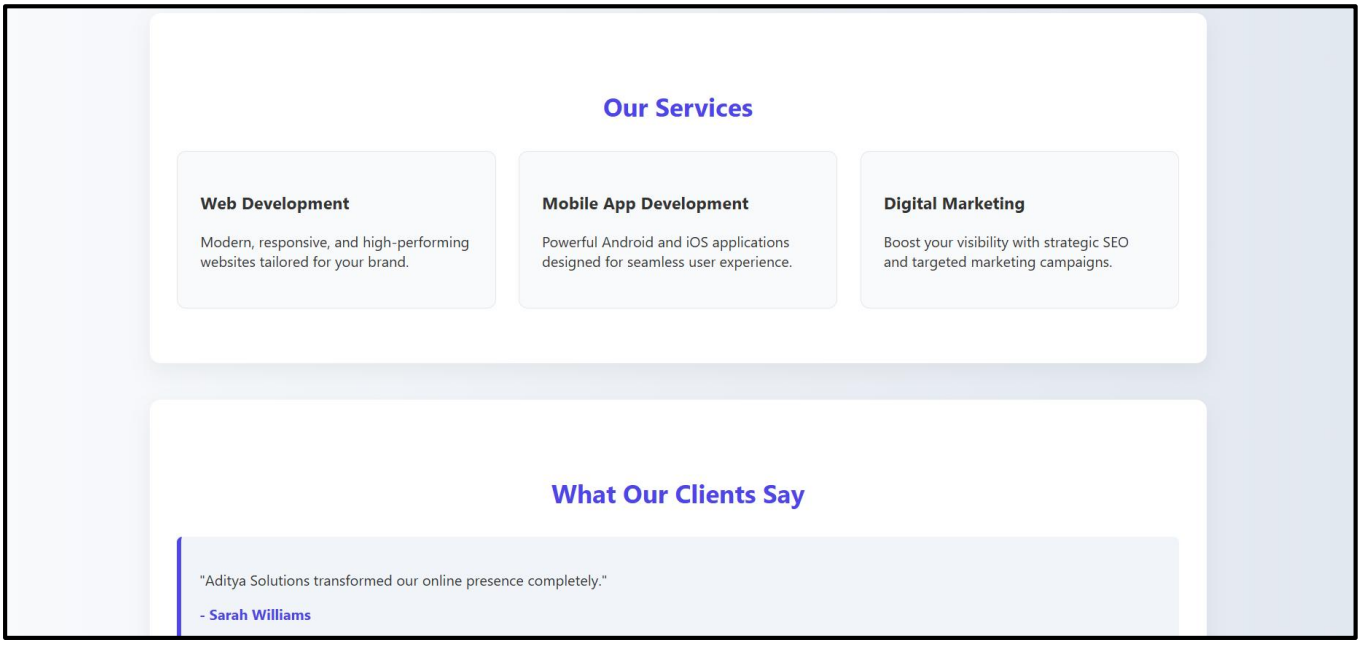
```
    background-color: white;
    color: #4f46e5;
    font-weight: bold;
    text-decoration: none;
    border-radius: 30px;
    transition: 0.3s ease;
}

.btn:hover {
    background-color: #e0e7ff;
    transform: translateY(-3px);
}

section {
    padding: 60px 30px;
    margin: 40px auto;
    max-width: 1100px;
    background-color: white;
    border-radius: 12px;
    box-shadow: 0 10px 25px rgba(0, 0, 0, 0.05);
}

section h2 {
    text-align: center;
    margin-bottom: 30px;
    font-size: 28px;
    color: #4f46e5;
}

.services-container {
    display: flex;
    gap: 25px;
    flex-wrap: wrap;
    justify-content: center;
}

.service-card {
    flex: 1;
    min-width: 260px;
    padding: 25px;
    border-radius: 10px;
    background: #f9fafb;
    transition: 0.3s ease;
    border: 1px solid #e5e7eb;
}

.service-card:hover {
    transform: translateY(-8px);
    box-shadow: 0 15px 25px rgba(0, 0, 0, 0.08);
}

.testimonial {
```

```
    padding: 20px;
    margin-bottom: 20px;
    background: #f1f5f9;
    border-left: 5px solid #4f46e5;
    border-radius: 6px;
}

.testimonial span {
    display: block;
    margin-top: 10px;
    font-weight: bold;
    color: #4f46e5;
}

.contact-info {
    text-align: center;
    margin-bottom: 25px;
}

.contact-form input,
.contact-form textarea {
    width: 100%;
    padding: 12px;
    margin-bottom: 15px;
    border-radius: 6px;
    border: 1px solid #d1d5db;
    font-size: 14px;
}

.contact-form input:focus,
.contact-form textarea:focus {
    outline: none;
    border-color: #4f46e5;
    box-shadow: 0 0 0 2px rgba(79, 70, 229, 0.2);
}

.contact-form button {
    background: linear-gradient(135deg, #4f46e5, #06b6d4);
    color: white;
    border: none;
    padding: 12px 25px;
    border-radius: 25px;
    cursor: pointer;
    font-size: 15px;
    transition: 0.3s ease;
}

.contact-form button:hover {
    opacity: 0.9;
    transform: translateY(-2px);
}
```

```
footer {
    text-align: center;
    padding: 20px;
    background: #1e293b;
    color: white;
    margin-top: 40px;
}
```

❖ OUTPUT:





"Exceptional service, creative ideas, and on-time delivery."

- James Anderson

Contact Us

Email: info@adityasolutions.com

Phone: +1 234 567 890

Address: 123 Innovation Street, Silicon City

Your Name

Your Email

Your Message

Contact Us

Email: info@adityasolutions.com

Phone: +1 234 567 890

Address: 123 Innovation Street, Silicon City

Your Name

Your Email

Your Message

Send Message

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EXPERIMENT – 2

❖ **AIM: Build a Company Landing Page using HTML and CSS.**

❖ **Tasks:**

- **Project Setup & Introduction –** Create a project folder with required files index.html and style.css
- **HTML Structure –** Building HTML structure including Hero Section, About section, Services, Testimonials, Contact Information.
- **Styling with CSS –** Appply CSS for Page layout, colors/fonts, alignment, margin/padding.
- **Service Section Design –** Adding atleast 3 services, including service name, short description
- **Testimonials –** Adding atleast 2 customer testimonials including customer name, and their feedback.
- **Contact Form –** Adding a contact form with address/email/phone

❖ **Codes:**

➢ **HTML Code:**

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Aditya Pal | Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <header class="hero">
        <div class="hero-content">
            <h1>Aditya Pal</h1>
            <p>B.Tech CSE Student</p>
            <p>K.R. Mangalam University | Second Year</p>
            <a href="#contact" class="btn">Contact Me</a>
```

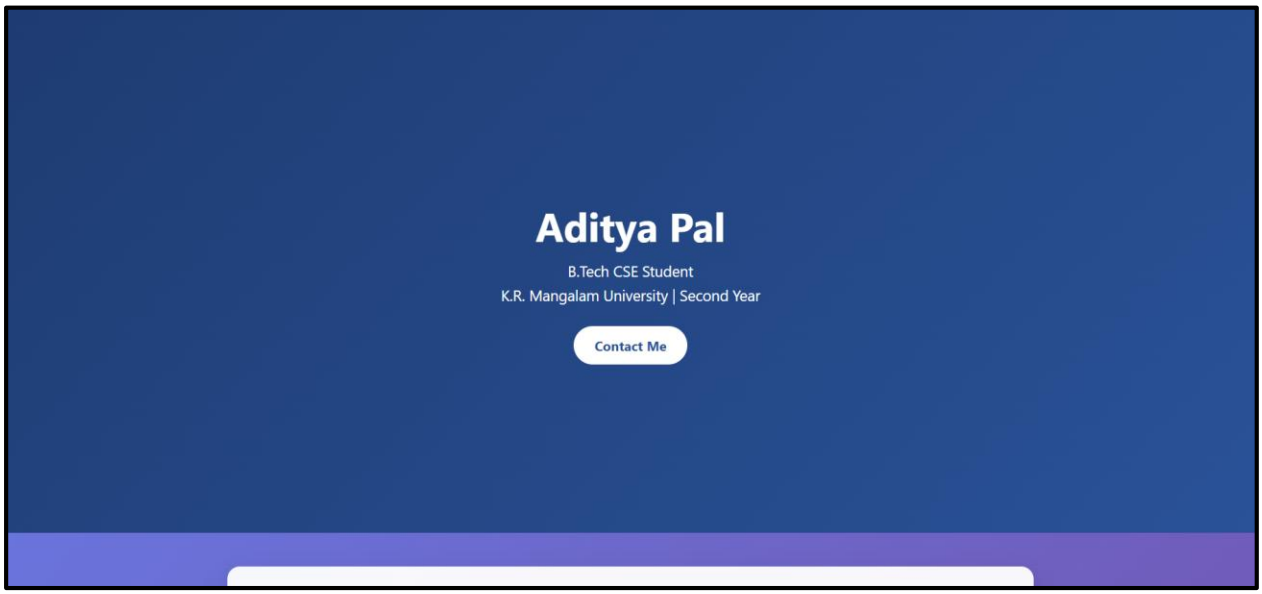
```html
        </div>
    </header>

    <section id="about" class="card">
        <h2>About Me</h2>
        <p>
            I am Aditya Pal, a second-year B.Tech Computer Science Engineering
student
            at K.R. Mangalam University. I am passionate about programming and
continuously
            learning new technologies to enhance my skills and build impactful
projects.
        </p>
    </section>

    <section id="skills" class="card">
        <h2>Skills</h2>
        <ul class="skills-list">
            <li>HTML</li>
            <li>CSS</li>
            <li>Python</li>
            <li>Java</li>
            <li>C++</li>
        </ul>
    </section>

    <section id="education" class="card">
        <h2>Education</h2>
        <p>
            <strong>Bachelor of Technology (B.Tech)</strong><br>
            Computer Science Engineering<br>
            K.R. Mangalam University<br>
            Currently in Second Year
        </p>
    </section>

    <section id="contact" class="card">
        <h2>Contact</h2>
        <p>Email: adityapal@example.com</p>
        <p>Phone: +91 9999999999</p>
    </section>

    <footer>
        <p>&copy; 2026 Aditya Pal. All rights reserved.</p>
    </footer>

</body>
</html>
```

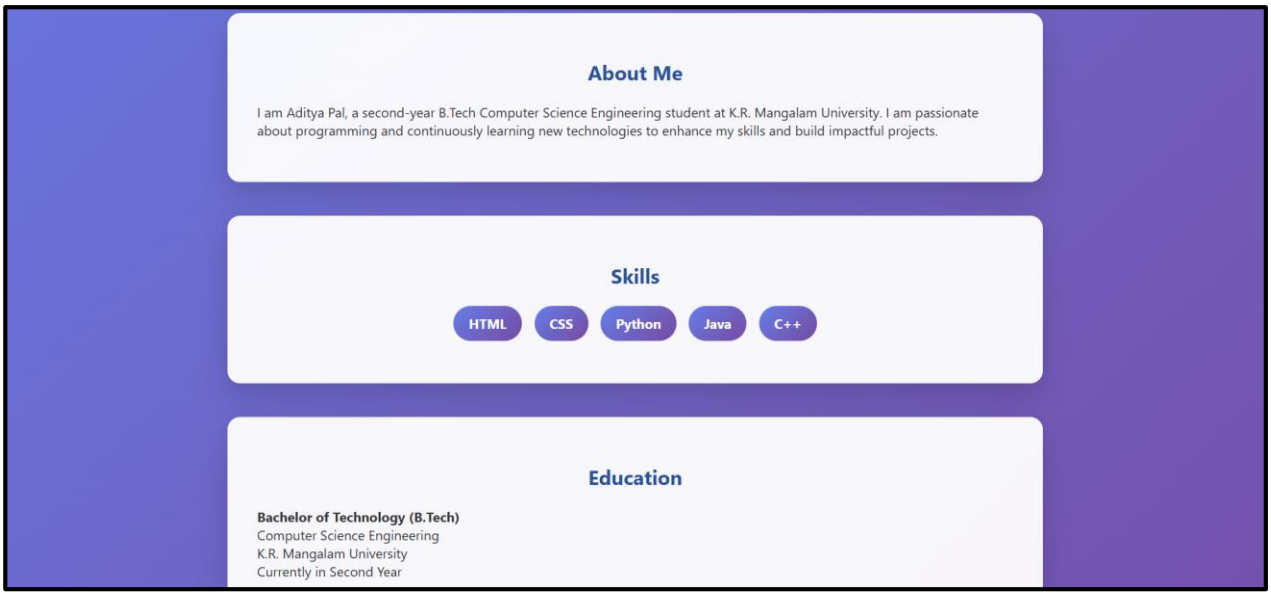
➢ **CSS Code:**

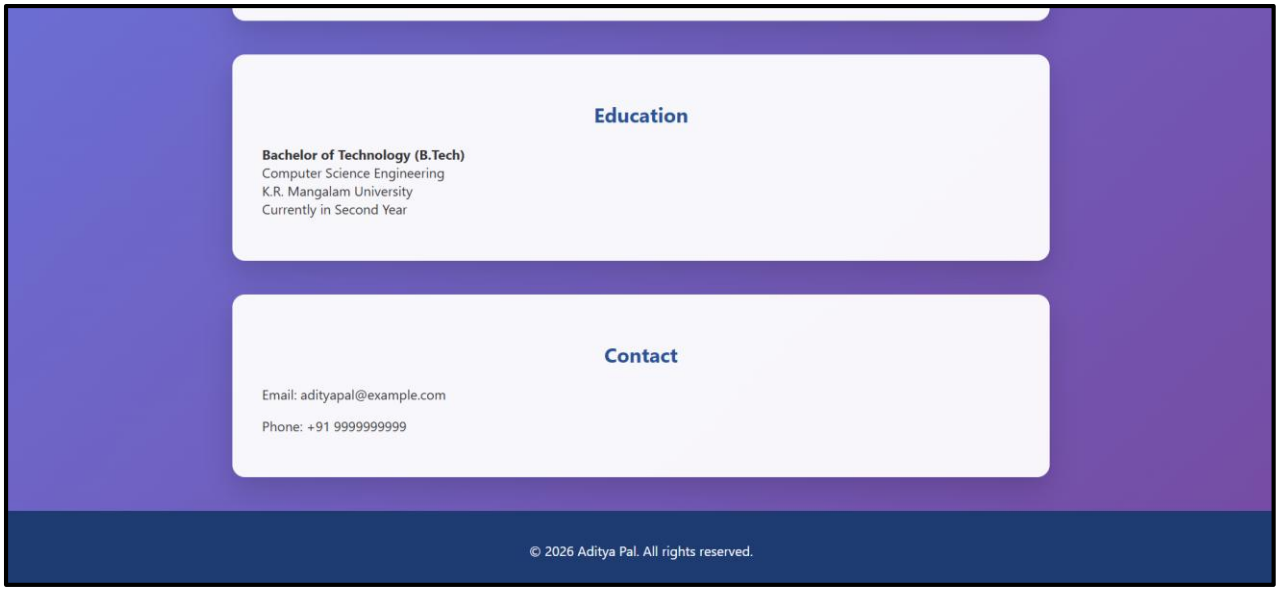
```
body {
    margin: 0;
    padding: 0;
    font-family: 'Segoe UI', sans-serif;
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: #333;
}

.hero {
    height: 90vh;
    display: flex;
    justify-content: center;
    align-items: center;
    text-align: center;
    color: white;
    background: linear-gradient(135deg, #1e3c72, #2a5298);
}

.hero-content h1 {
    font-size: 48px;
    margin-bottom: 10px;
}

.hero-content p {
    font-size: 18px;
    margin: 5px 0;
}

.btn {
    display: inline-block;
    margin-top: 20px;
    padding: 12px 25px;
    background: white;
    color: #2a5298;
    border-radius: 30px;
    text-decoration: none;
    font-weight: bold;
    transition: 0.3s ease;
}

.btn:hover {
    background: #e0e7ff;
    transform: translateY(-3px);
}

.card {
    max-width: 900px;
    margin: 40px auto;
    padding: 35px;
    border-radius: 15px;
    background: rgba(255, 255, 255, 0.95);
    box-shadow: 0 15px 35px rgba(0, 0, 0, 0.15);
```

```
}

.card h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #2a5298;
}

.skills-list {
    list-style: none;
    padding: 0;
    display: flex;
    gap: 15px;
    flex-wrap: wrap;
    justify-content: center;
}

.skills-list li {
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    padding: 10px 18px;
    border-radius: 25px;
    font-weight: bold;
    transition: 0.3s ease;
}

.skills-list li:hover {
    transform: scale(1.1);
    cursor: pointer;
}

footer {
    text-align: center;
    padding: 20px;
    background: #1e3c72;
    color: white;
    margin-top: 40px;
}
```

❖ OUTPUT:


Aditya Pal
B.Tech CSE Student
K.R. Mangalam University | Second Year
Contact Me


About Me
I am Aditya Pal, a second-year B.Tech Computer Science Engineering student at K.R. Mangalam University. I am passionate about programming and continuously learning new technologies to enhance my skills and build impactful projects.
Skills
HTML
CSS
Python
Java
C++
Education
Bachelor of Technology (B.Tech)
Computer Science Engineering
K.R. Mangalam University
Currently in Second Year


Education
Bachelor of Technology (B.Tech)
Computer Science Engineering
K.R. Mangalam University
Currently in Second Year
Contact
Email: adityapal@example.com
Phone: +91 9999999999
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EXPERIMENT-3

❖ **AIM: Build an Interactive Quiz Application using HTML, CSS, and JavaScript.**

❖ **Tasks:**

- **Project Setup & Introduction –** Create a project folder named interactive_quiz with required files: index.html, style.css, and script.js. Add header comments in HTML (project title, name, date) and display a brief quiz introduction.

- **HTML Structure –** Build a structured quiz interface including question text, multiple-choice options (radio buttons or buttons), and a "Next" or "Submit" button using semantic HTML elements.

- **Styling with CSS –** Apply CSS to improve readability, align buttons and sections properly, and use appropriate colors and spacing for better user experience.

- **JavaScript Logic & Interactivity –** Store quiz questions and answers in JavaScript (array/object), capture user-selected answers, navigate between questions using button events, and track/update the score dynamically.

- **Result Display –** Show final score after quiz completion along with a feedback message such as "Excellent", "Good", or "Try Again". Optionally hide the quiz interface after submission.

- **Restart Quiz Feature (Bonus) –** Add a "Restart Quiz" button to reset score and question index, making the application reusable.

❖ Codes:

➢ HTML Code:

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Interactive Quiz | Aditya Pal</title>

    <!-- Google Font -->
    <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700&display=
swap" rel="stylesheet">

    <link rel="stylesheet" href="style.css">
</head>

<body>

    <main class="quiz-container">

        <section id="intro">
            <h1>Interactive Quiz</h1>
            <p>Test your programming and web development basics</p>
        </section>

        <div class="progress">
            <div id="progress-bar"></div>
        </div>

        <section id="quiz">
            <h2 id="question"></h2>

            <div class="options">
                <label class="option-item">
                    <input type="radio" name="option" value="0">
                    <span id="opt0"></span>
                </label>

                <label class="option-item">
                    <input type="radio" name="option" value="1">
                    <span id="opt1"></span>
                </label>

                <label class="option-item">
                    <input type="radio" name="option" value="2">
```

```html
                <span id="opt2"></span>
            </label>

            <label class="option-item">
                <input type="radio" name="option" value="3">
                <span id="opt3"></span>
            </label>
        </div>

        <button id="nextBtn">Next Question</button>
    </section>

    <section id="result" class="hidden">
        <h2>Quiz Result</h2>
        <p id="scoreText"></p>
        <p id="feedback"></p>

        <div id="review"></div>

        <button id="restartBtn">Restart Quiz</button>
    </section>

  </main>

  <script src="script.js"></script>
</body>

</html>
```

➢ CSS Code:

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: "Poppins", "Segoe UI", sans-serif;
}

body {
    min-height: 100vh;
    background: linear-gradient(135deg, #667eea, #764ba2, #6a11cb);
    background-size: 300% 300%;
    animation: gradientMove 8s ease infinite;
    display: flex;
    justify-content: center;
    align-items: center;
}

@keyframes gradientMove {
    0% { background-position: 0% 50%; }
```

```
    50% { background-position: 100% 50%; }
    100% { background-position: 0% 50%; }
}

.quiz-container {
    width: 520px;
    max-width: 95%;
    padding: 35px;
    border-radius: 20px;
    background: rgba(255, 255, 255, 0.12);
    backdrop-filter: blur(18px);
    -webkit-backdrop-filter: blur(18px);
    border: 1px solid rgba(255, 255, 255, 0.25);
    box-shadow: 0 25px 50px rgba(0, 0, 0, 0.25);
    animation: fadeIn 0.6s ease-in-out;
    color: white;
}

@keyframes fadeIn {
    from {
        opacity: 0;
        transform: translateY(25px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

#intro {
    text-align: center;
    margin-bottom: 25px;
}

#intro h1 {
    font-size: 30px;
    margin-bottom: 8px;
    letter-spacing: 1px;
}

#intro p {
    font-size: 14px;
    opacity: 0.85;
}

.progress {
    width: 100%;
    height: 10px;
    background: rgba(255, 255, 255, 0.2);
    border-radius: 10px;
    overflow: hidden;
    margin-bottom: 25px;
```

```
}

#progress-bar {
    height: 100%;
    width: 0%;
    background: linear-gradient(90deg, #00f5a0, #00d9f5);
    box-shadow: 0 0 15px rgba(0, 255, 200, 0.8);
    transition: width 0.4s ease;
}

#question {
    font-size: 18px;
    margin-bottom: 18px;
    font-weight: 500;
}

.options label {
    display: flex;
    align-items: center;
    padding: 14px;
    margin-bottom: 12px;
    border-radius: 12px;
    background: rgba(255, 255, 255, 0.15);
    cursor: pointer;
    transition: all 0.3s ease;
    border: 1px solid transparent;
}

.options label:hover {
    background: rgba(255, 255, 255, 0.25);
    transform: translateY(-2px);
    border: 1px solid rgba(255, 255, 255, 0.4);
}

.options input {
    margin-right: 10px;
    accent-color: #00f5a0;
    cursor: pointer;
}

button {
    width: 100%;
    padding: 14px;
    border-radius: 12px;
    border: none;
    background: linear-gradient(135deg, #00f5a0, #00d9f5);
    color: #111;
    font-size: 16px;
    font-weight: 600;
    cursor: pointer;
    margin-top: 18px;
    transition: all 0.3s ease;
```

```
}

button:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 20px rgba(0, 255, 200, 0.5);
}

.hidden {
    display: none;
}

#result {
    text-align: center;
}

#result h2 {
    margin-bottom: 12px;
}

#scoreText {
    font-size: 20px;
    margin-bottom: 5px;
}

#feedback {
    font-weight: bold;
    font-size: 16px;
    margin-bottom: 20px;
    color: #00f5a0;
}

.review-item {
    text-align: left;
    background: rgba(255, 255, 255, 0.15);
    padding: 14px;
    border-radius: 12px;
    margin-bottom: 12px;
    border-left: 4px solid #00f5a0;
}

.review-item p {
    font-size: 14px;
    margin-bottom: 5px;
}

.correct {
    color: #00ff99;
    font-weight: 600;
}

.wrong {
    color: #ff6b6b;
```

```
    font-weight: 600;
}

@media (max-width: 500px) {
    .quiz-container {
        padding: 25px;
    }

    #intro h1 {
        font-size: 24px;
    }

    #question {
        font-size: 16px;
    }
}
```

➢ JAVASCRIPT Code:

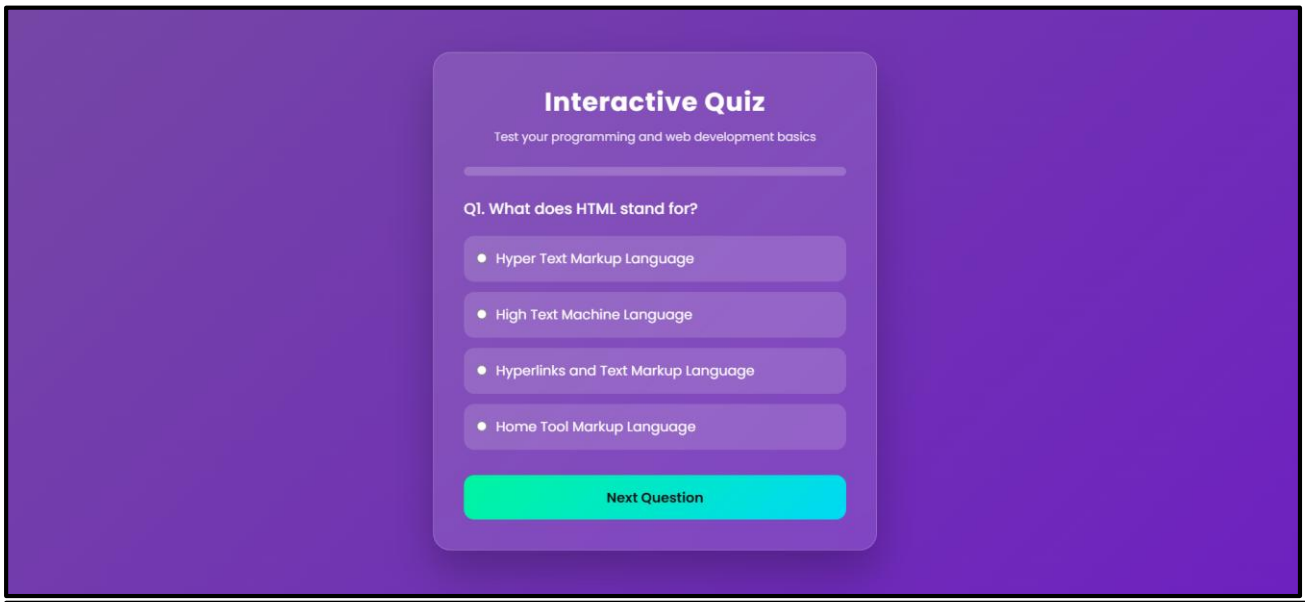
```
const quizData = [
    {
        question: "What does HTML stand for?",
        options: [
            "Hyper Text Markup Language",
            "High Text Machine Language",
            "Hyperlinks and Text Markup Language",
            "Home Tool Markup Language"
        ],
        answer: 0
    },
    {
        question: "Which language is used for styling web pages?",
        options: ["HTML", "Java", "CSS", "Python"],
        answer: 2
    },
    {
        question: "Which tag creates a hyperlink?",
        options: ["<link>", "<a>", "<href>", "<url>"],
        answer: 1
    },
    {
        question: "Which language runs in the browser?",
        options: ["Java", "C++", "Python", "JavaScript"],
        answer: 3
    },
    {
        question: "CSS comments use which syntax?",
        options: ["//", "#", "/* */", "<!-- -->"],
        answer: 2
    },
```

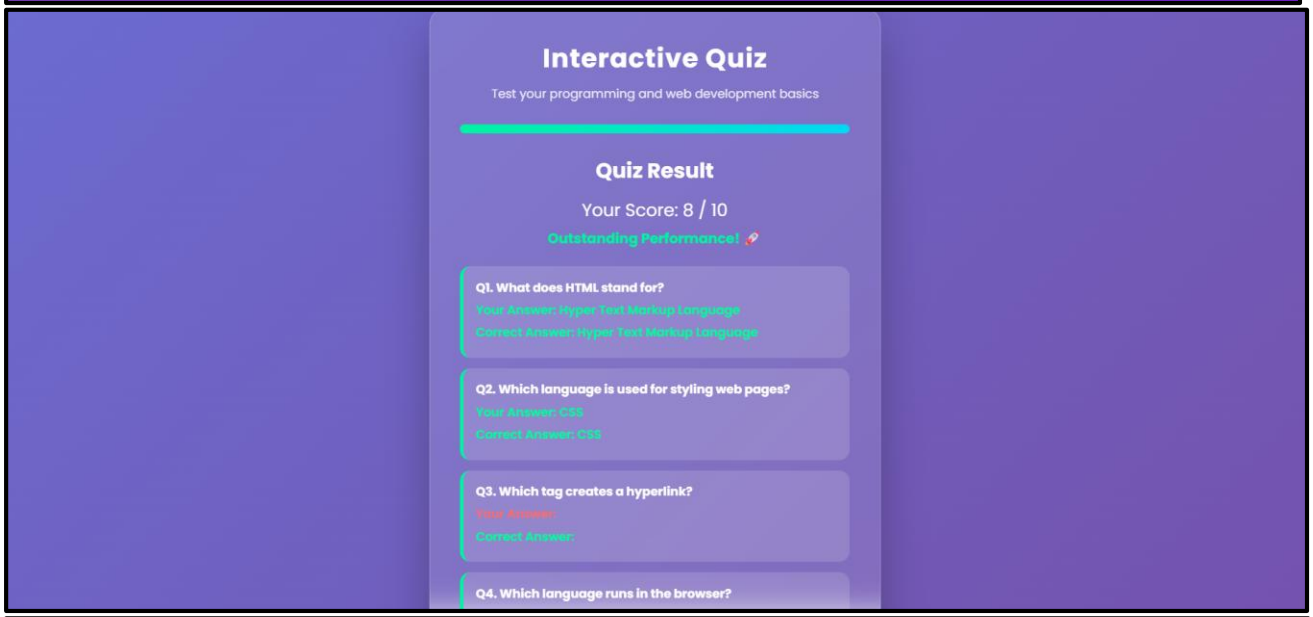
```
    {
        question: "Which keyword declares a variable in JS?",
        options: ["var", "int", "string", "define"],
        answer: 0
    },
    {
        question: "Which is a semantic HTML tag?",
        options: ["<div>", "<span>", "<section>", "<b>"],
        answer: 2
    },
    {
        question: "CSS property for text size?",
        options: ["font-style", "text-size", "font-size", "size"],
        answer: 2
    },
    {
        question: "Java was developed by?",
        options: ["Microsoft", "Sun Microsystems", "Google", "Apple"],
        answer: 1
    },
    {
        question: "Function keyword in JavaScript?",
        options: ["method", "function", "def", "func"],
        answer: 1
    }
];

let currentQuestion = 0;
let score = 0;
let userAnswers = [];

const questionEl = document.getElementById("question");
const options = document.querySelectorAll("input[name='option']");
const nextBtn = document.getElementById("nextBtn");
const quizSection = document.getElementById("quiz");
const resultSection = document.getElementById("result");
const scoreText = document.getElementById("scoreText");
const feedback = document.getElementById("feedback");
const restartBtn = document.getElementById("restartBtn");
const reviewDiv = document.getElementById("review");
const progressBar = document.getElementById("progress-bar");

function loadQuestion() {
    const q = quizData[currentQuestion];

    questionEl.textContent = `Q${currentQuestion + 1}. ${q.question}`;

    q.options.forEach((opt, i) => {
        document.getElementById("opt" + i).textContent = opt;
        options[i].checked = false;
    });
```

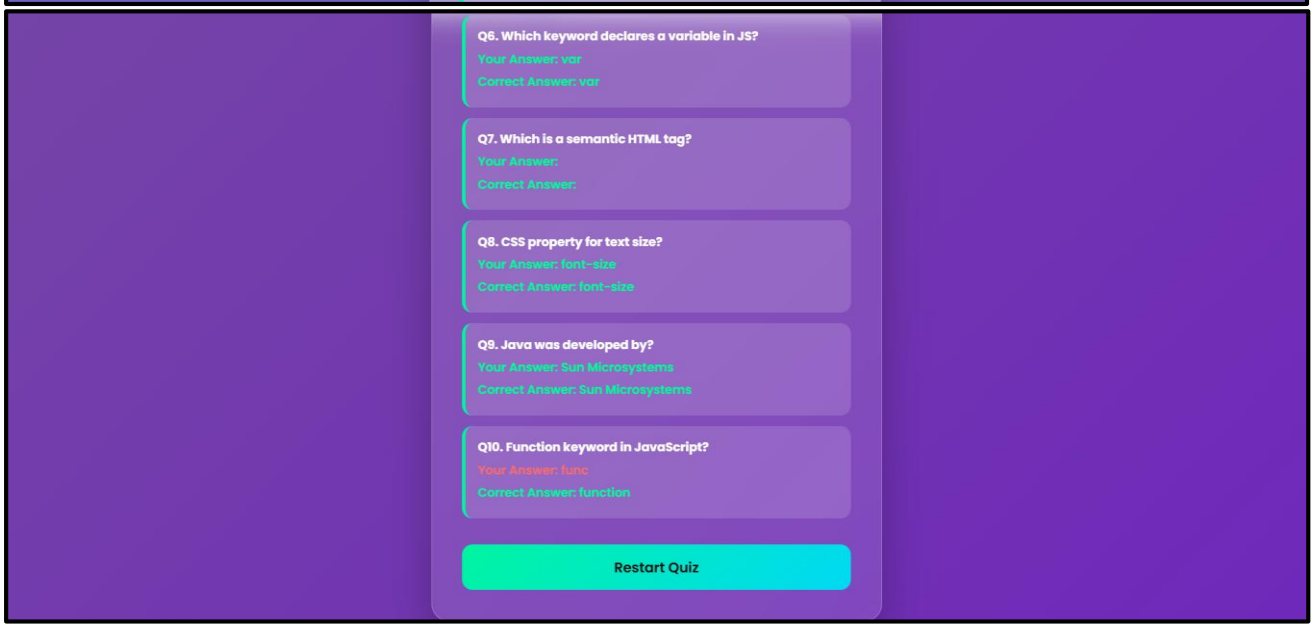
```
    progressBar.style.width = `${(currentQuestion / quizData.length) * 100}%`;

    nextBtn.textContent =
        currentQuestion === quizData.length - 1
            ? "Submit Quiz"
            : "Next Question";
}

function getSelected() {
    let selected = null;
    options.forEach(opt => {
        if (opt.checked) {
            selected = parseInt(opt.value);
        }
    });
    return selected;
}

nextBtn.addEventListener("click", () => {
    const selected = getSelected();

    if (selected === null) {
        alert("Please select an option before continuing.");
        return;
    }

    userAnswers.push(selected);

    if (selected === quizData[currentQuestion].answer) {
        score++;
    }

    currentQuestion++;

    if (currentQuestion < quizData.length) {
        loadQuestion();
    } else {
        showResult();
    }
});

function showResult() {
    quizSection.classList.add("hidden");
    resultSection.classList.remove("hidden");
    progressBar.style.width = "100%";

    scoreText.textContent = `Your Score: ${score} / ${quizData.length}`;

    if (score >= 8) {
        feedback.textContent = "Outstanding Performance! 🚀";
    } else if (score >= 5) {
        feedback.textContent = "Great Job! Keep Improving 👍";
```

```
    } else {
        feedback.textContent = "Keep Practicing 💡";
    }

    reviewDiv.innerHTML = "";

    quizData.forEach((q, i) => {
        const div = document.createElement("div");
        div.classList.add("review-item");

        div.innerHTML = `
            <p><strong>Q${i + 1}. ${q.question}</strong></p>
            <p class="${userAnswers[i] === q.answer ? "correct" : "wrong"}">
                Your Answer: ${q.options[userAnswers[i]]}
            </p>
            <p class="correct">
                Correct Answer: ${q.options[q.answer]}
            </p>
        `;

        reviewDiv.appendChild(div);
    });
}

restartBtn.addEventListener("click", () => {
    currentQuestion = 0;
    score = 0;
    userAnswers = [];
    resultSection.classList.add("hidden");
    quizSection.classList.remove("hidden");
    loadQuestion();
});

loadQuestion();
```

❖ OUTPUT:


Interactive Quiz
Test your programming and web development basics
Q1. What does HTML stand for?
Hyper Text Markup Language
High Text Machine Language
Hyperlinks and Text Markup Language
Home Tool Markup Language
Next Question


Interactive Quiz
Test your programming and web development basics
Quiz Result
Your Score: 8 / 10
Outstanding Performance! 🚀
Q1. What does HTML stand for?
Your Answer: Hyper Text Markup Language
Correct Answer: Hyper Text Markup Language
Q2. Which language is used for styling web pages?
Your Answer: CSS
Correct Answer: CSS
Q3. Which tag creates a hyperlink?
Your Answer:
Correct Answer:
Q4. Which language runs in the browser?


Q6. Which keyword declares a variable in JS?
Your Answer: var
Correct Answer: var
Q7. Which is a semantic HTML tag?
Your Answer:
Correct Answer:
Q8. CSS property for text size?
Your Answer: font-size
Correct Answer: font-size
Q9. Java was developed by?
Your Answer: Sun Microsystems
Correct Answer: Sun Microsystems
Q10. Function keyword in JavaScript?
Your Answer: func
Correct Answer: function
Restart Quiz

EXPERIMENT-4

❖ **AIM: Build a Dynamic To-Do List Application using HTML, CSS, and JavaScript.**

❖ **Tasks:**

• **Project Setup & Introduction –** Create a project folder named dynamic_todo_list containing the required files: index.html, style.css, and script.js. Add header comments in the HTML file (project title, name, date) and display a short description of the to-do list application.

• **To-Do List Interface (HTML) –** Design the basic interface using HTML. Create an input field for entering tasks, an "Add Task" button, and display tasks in a list format using <ul> or <ol> elements.

• **Styling the To-Do List (CSS) –** Apply CSS styling to improve the interface appearance. Style the input box and buttons, align the task list properly, and visually separate tasks to keep the layout clean and readable.

• **Add & Display Tasks (JavaScript) –** Use JavaScript to capture user input and dynamically add new tasks to the list. Implement validation to prevent empty tasks from being added.

• **Edit & Delete Tasks –** Add "Edit" and "Delete" buttons for each task. Allow users to modify the task text and remove tasks dynamically from the list.

• **Task Completion Status (Bonus) –** Add a checkbox or toggle option to mark tasks as completed. Display completed tasks with a visual indicator such as strikethrough text, and optionally show the count of completed vs pending tasks**.**

- **Submission Instructions:**
  - Submit the dynamic_todo_list project folder.
  - Include index.html, style.css, and script.js files.
  - Provide at least 2 screenshots (before and after task updates).
  - Ensure the code is clean and well-commented.
  - Submit the GitHub repository URL via LMS.

❖ Codes:

➢ HTML Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Smart To-Do List</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="app-container">

    <h1>✨ To-Do List</h1>
    <p class="subtitle">Stay organized. Stay productive.</p>


    <div class="input-group">
        <input type="text" id="taskInput" placeholder="What needs to be done?">
        <button id="addTaskBtn">+</button>
    </div>


    <div class="stats">
        <span id="taskCount">0 Tasks</span>
        <span id="completedCount">0 Completed</span>
    </div>


    <div class="progress-container">
        <div id="progressBar"></div>
    </div>


    <ul id="taskList"></ul>


    <button id="clearAllBtn" class="clear-btn">Clear All</button>
```

```
</div>

<script src="script.js"></script>
</body>
</html>
```

➢ CSS Code:

```
* {
    box-sizing: border-box;
}

body {
    margin: 0;
    font-family: "Segoe UI", Tahoma, sans-serif;
    background-color: #f4f6f8;
    display: flex;
    justify-content: center;
    padding: 40px 15px;
}


.app-container {
    width: 100%;
    max-width: 520px;
    background-color: #ffffff;
    padding: 30px;
    border-radius: 8px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
}


h1 {
    margin: 0;
    font-size: 24px;
    font-weight: 600;
    color: #1f2937;
    text-align: center;
}

.subtitle {
    text-align: center;
    font-size: 14px;
    color: #6b7280;
    margin: 8px 0 25px;
}


.input-group {
```

```
    display: flex;
}

#taskInput {
    flex: 1;
    padding: 10px 12px;
    border: 1px solid #d1d5db;
    border-radius: 6px 0 0 6px;
    font-size: 14px;
    outline: none;
}

#taskInput:focus {
    border-color: #2563eb;
}

#addTaskBtn {
    width: 50px;
    border: none;
    background-color: #2563eb;
    color: white;
    font-size: 18px;
    border-radius: 0 6px 6px 0;
    cursor: pointer;
    transition: background 0.2s ease;
}

#addTaskBtn:hover {
    background-color: #1e4fd8;
}


.stats {
    display: flex;
    justify-content: space-between;
    margin: 18px 0 8px;
    font-size: 13px;
    color: #4b5563;
}


.progress-container {
    width: 100%;
    height: 6px;
    background-color: #e5e7eb;
    border-radius: 4px;
    margin-bottom: 20px;
}

#progressBar {
    height: 100%;
    width: 0%;
```

```
    background-color: #2563eb;
    border-radius: 4px;
    transition: width 0.3s ease;
}


ul {
    list-style: none;
    padding: 0;
    margin: 0;
}

li {
    display: flex;
    align-items: center;
    justify-content: space-between;
    background-color: #f9fafb;
    padding: 12px;
    margin-bottom: 10px;
    border: 1px solid #e5e7eb;
    border-radius: 6px;
    transition: background 0.2s ease;
}

li:hover {
    background-color: #f3f4f6;
}


.task-info {
    flex: 1;
    margin-left: 10px;
    font-size: 14px;
    color: #111827;
}

.task-info small {
    color: #6b7280;
    font-size: 12px;
}


.completed {
    text-decoration: line-through;
    color: #9ca3af;
}


.action-btn {
    border: none;
    padding: 6px 10px;
    margin-left: 6px;
```

```
    border-radius: 4px;
    font-size: 12px;
    cursor: pointer;
    transition: background 0.2s ease;
}


.edit {
    background-color: #e5e7eb;
    color: #111827;
}

.edit:hover {
    background-color: #d1d5db;
}


.delete {
    background-color: #ef4444;
    color: white;
}

.delete:hover {
    background-color: #dc2626;
}


.clear-btn {
    width: 100%;
    margin-top: 18px;
    padding: 10px;
    border: none;
    border-radius: 6px;
    background-color: #374151;
    color: white;
    font-size: 14px;
    cursor: pointer;
    transition: background 0.2s ease;
}

.clear-btn:hover {
    background-color: #1f2937;
}


@media (max-width: 480px) {
    .app-container {
        padding: 20px;
    }

    h1 {
        font-size: 20px;
```

```
    }
}
```

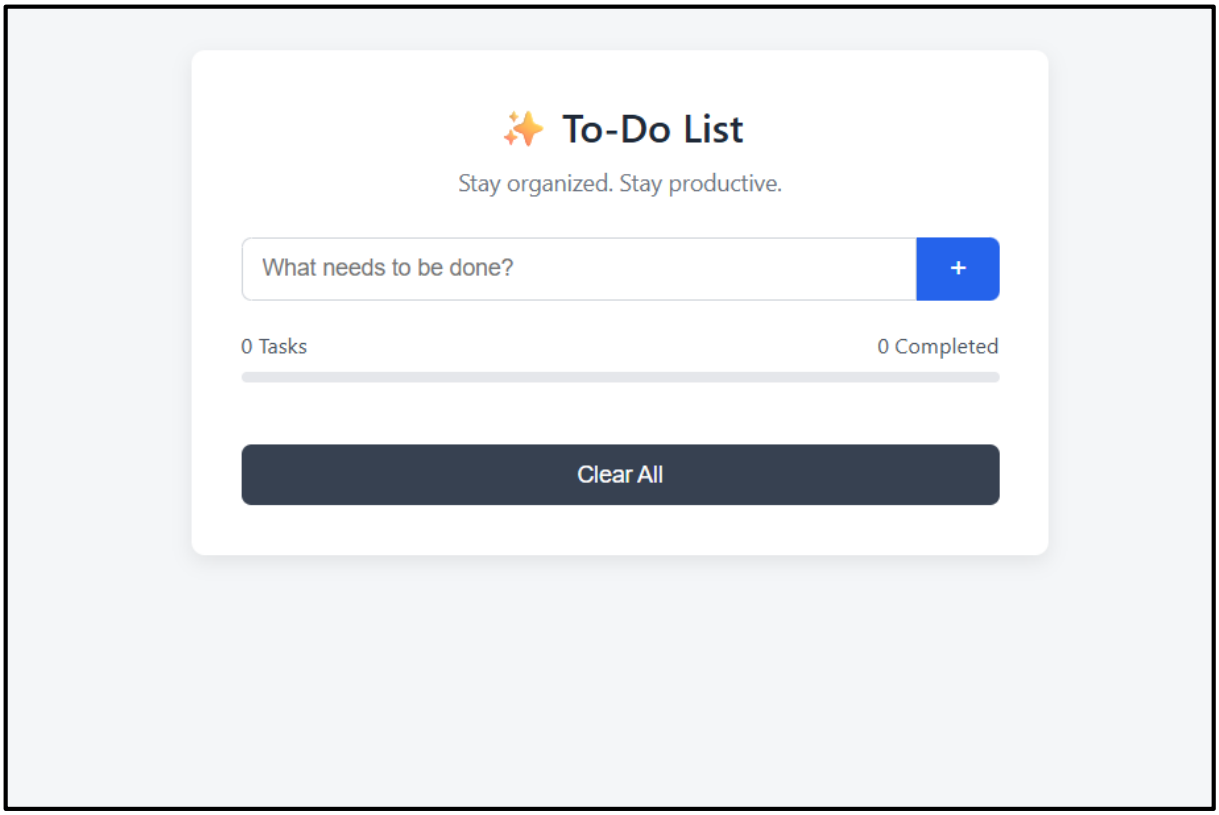
➢ JAVASCRIPT Code:

```
const taskInput = document.getElementById("taskInput");
const addTaskBtn = document.getElementById("addTaskBtn");
const taskList = document.getElementById("taskList");
const taskCount = document.getElementById("taskCount");
const completedCount = document.getElementById("completedCount");
const progressBar = document.getElementById("progressBar");
const clearAllBtn = document.getElementById("clearAllBtn");

let tasks = JSON.parse(localStorage.getItem("tasks")) || [];


function addTask(text) {
    const task = {
        id: Date.now(),
        text: text,
        completed: false,
        date: new Date().toLocaleString()
    };
    tasks.push(task);
    saveTasks();
    renderTasks();
}


function renderTasks() {
    taskList.innerHTML = "";
    let completedTasks = 0;

    tasks.forEach(task => {
        const li = document.createElement("li");

        const checkbox = document.createElement("input");
        checkbox.type = "checkbox";
        checkbox.checked = task.completed;

        const div = document.createElement("div");
        div.className = "task-info";
        div.innerHTML = `<strong>${task.text}</strong><br><small>${task.date}</small>`;

        if (task.completed) {
            div.classList.add("completed");
            completedTasks++;
        }

        checkbox.addEventListener("change", () => {
```

```
            task.completed = checkbox.checked;
            saveTasks();
            renderTasks();
        });

        const editBtn = document.createElement("button");
        editBtn.textContent = "Edit";
        editBtn.className = "action-btn edit";
        editBtn.onclick = () => {
            const newText = prompt("Edit task:", task.text);
            if (newText && newText.trim() !== "") {
                task.text = newText.trim();
                saveTasks();
                renderTasks();
            }
        };

        const deleteBtn = document.createElement("button");
        deleteBtn.textContent = "Delete";
        deleteBtn.className = "action-btn delete";
        deleteBtn.onclick = () => {
            tasks = tasks.filter(t => t.id !== task.id);
            saveTasks();
            renderTasks();
        };

        li.appendChild(checkbox);
        li.appendChild(div);
        li.appendChild(editBtn);
        li.appendChild(deleteBtn);

        taskList.appendChild(li);
    });

    updateStats(completedTasks);
}


function updateStats(completedTasks) {
    taskCount.textContent = `${tasks.length} Tasks`;
    completedCount.textContent = `${completedTasks} Completed`;

    const percent = tasks.length === 0 ? 0 :
        (completedTasks / tasks.length) * 100;
    progressBar.style.width = percent + "%";
}


function saveTasks() {
    localStorage.setItem("tasks", JSON.stringify(tasks));
}
```

```
addTaskBtn.addEventListener("click", () => {
    const text = taskInput.value.trim();
    if (text !== "") {
        addTask(text);
        taskInput.value = "";
    }
});


taskInput.addEventListener("keypress", (e) => {
    if (e.key === "Enter") {
        addTaskBtn.click();
    }
});


clearAllBtn.addEventListener("click", () => {
    if (confirm("Are you sure you want to delete all tasks?")) {
        tasks = [];
        saveTasks();
        renderTasks();
    }
});

renderTasks();
```

❖ OUTPUT-



THANK
YOU!